INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE

Van-Con School Bus Builders

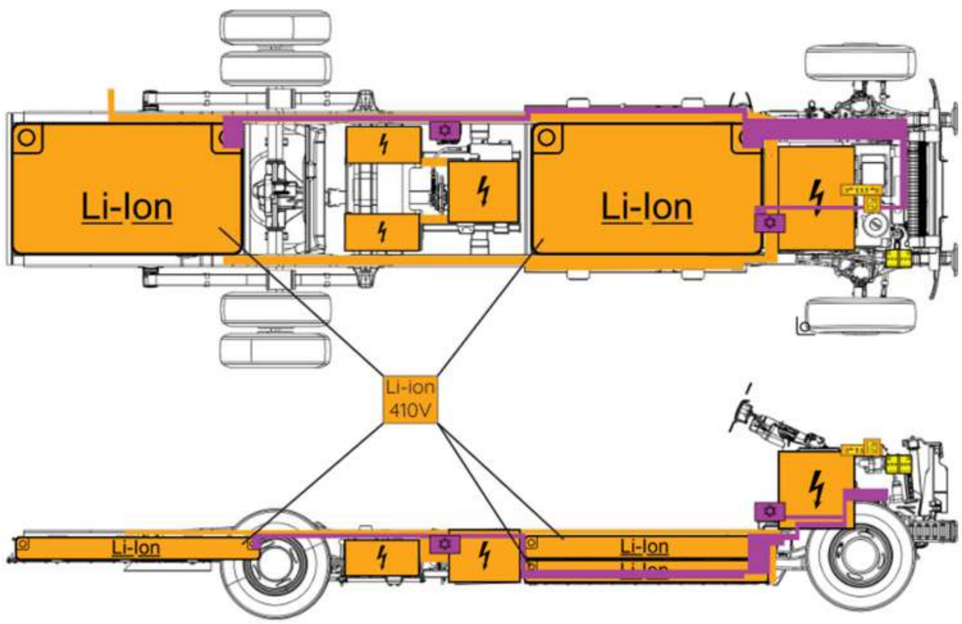
Electric School Bus

ELECTRIC

CONTENTS

0. Rescue Sheet

VAN-CON School Bus Builders	Electric School Bus 2025	



High-Voltage Battery	High-Voltage Component	High-Voltage Cable / Component	Emergency Disconnect Switch
HV Cable Cut	12V Battery	Air Conditioning Component	Air Conditioning Line

1. Identification / recognition

Van-Con electric buses are built on Motiv Ford F-450 frame and Motiv installed components.

Vehicle Badging

Van-Con badging is located on both sides of the cab or along the length of the body and the rear of the bus as pictured below.

Vehicle Labeling

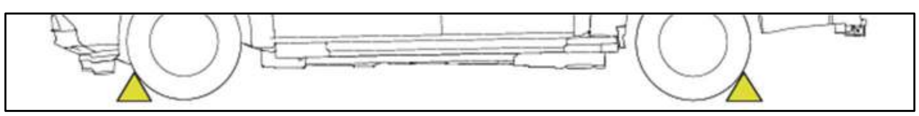
Van-Con vehicles can be identified by the Intermediate Vehicle Manufacturer (IVM) label applied to the driver side-door frame.

Under the Hood

Van-Con vehicles can be identified by the Motiv electronics package and Motiv logo that may be visible on components under the hood.

2. Immobilization / stabilization / lifting

If it is necessary to park the vehicle follow these steps:

Step	Description
1	Turn the key switch to the **ON** position.
2	Apply and hold the brake pedal.
3	Press the **P** (park button on the shift mode selector). 
4	Apply the parking brake.
5	Turn the key switch to the **OFF** position and remove the key.
6	Chock the wheels. 

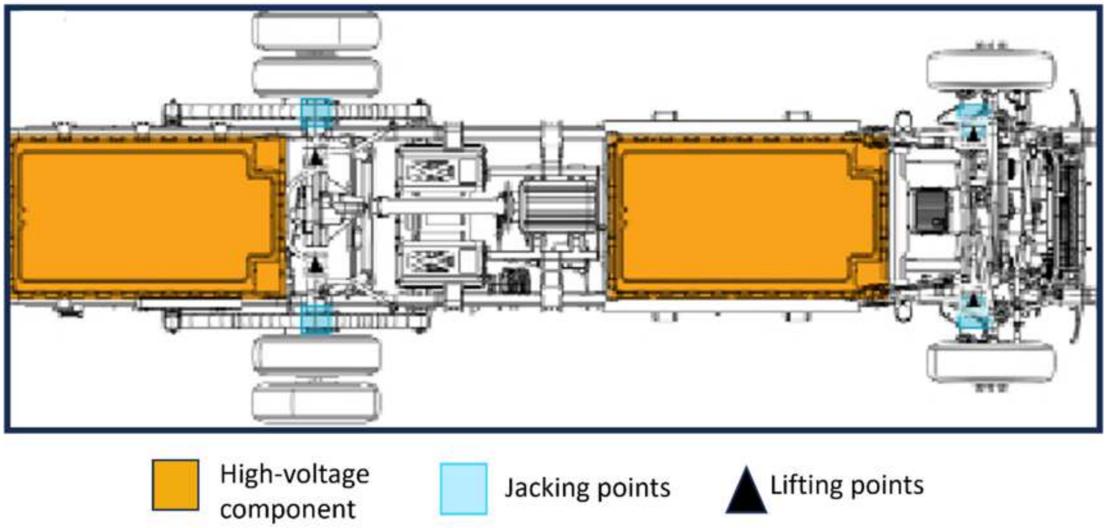
Stabilization/Lift Points

WARNING: Use caution to avoid damage to the high-voltage battery while stabilizing/lifting the vehicle.

WARNING: Do not use the high-voltage battery to lift or stabilize the vehicle.

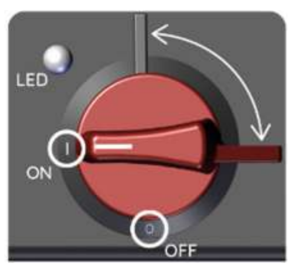
WARNING: Use caution to avoid coming in contact with the high-voltage battery or other high-voltage component.

WARNING: Only first responders who are trained and equipped at the technician level per the National Fire Protection Association (NFPA) and are familiar with the lift points should be lifting or manipulating the vehicle.



3. Disable direct hazards / safety regulations

Emergency Cut-off Switch



The red emergency cut-off switch is the main power switch and is located under the hood. This switch disconnects the 12V supply and the high-voltage system.

When this happens, the high-voltage battery controllers disconnect the batteries internally, isolating high-voltage safely within the batteries.

The high-voltage cable can be cut during emergencies. It is located under the hood.

WARNING: Do not touch, cut, or open high-voltage components and batteries.

WARNING: Use caution to avoid damaging the high-voltage batteries during vehicle rescue and recovery operations.

WARNING: There is no way to instantly discharge the energy that is stored inside of the high-voltage batteries.

The 12V electrical system provides power to all essential and accessory vehicle services (standard lights, fans, communication equipment, etc.). The 12V battery is located under the hood.

Disconnect the Charging Connector

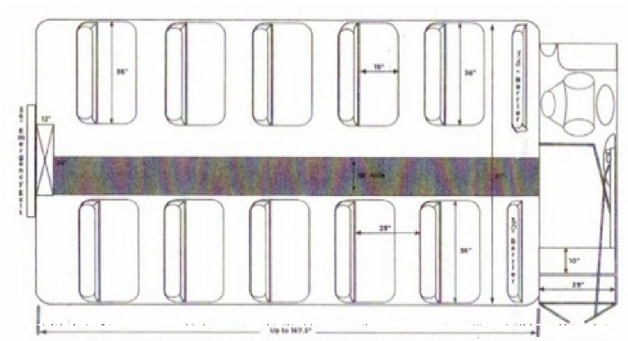
In the event of a vehicle fire or other situation that requires relocating the vehicle, make sure that the vehicle charging connector is disconnected before the vehicle is moved.

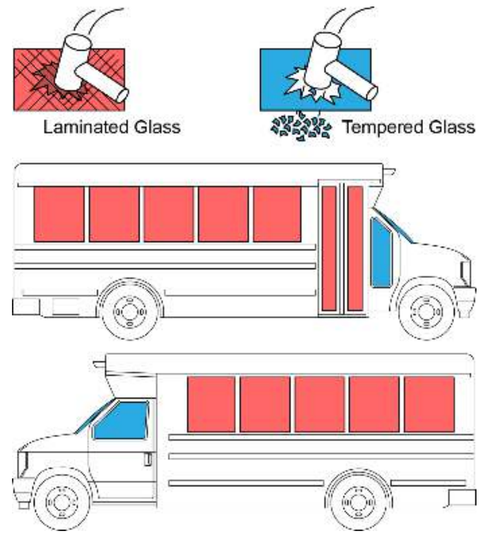
Electrical Box

The electrical box is located above the driver's area and contains fuses and cables for 12V components.

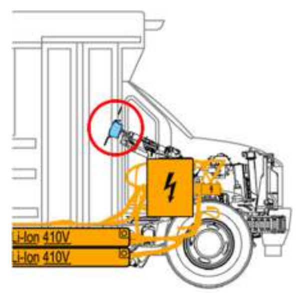
4. Access to the Occupants

The bus is configured in a 2-person seat with 2 seats in a row as shown below.



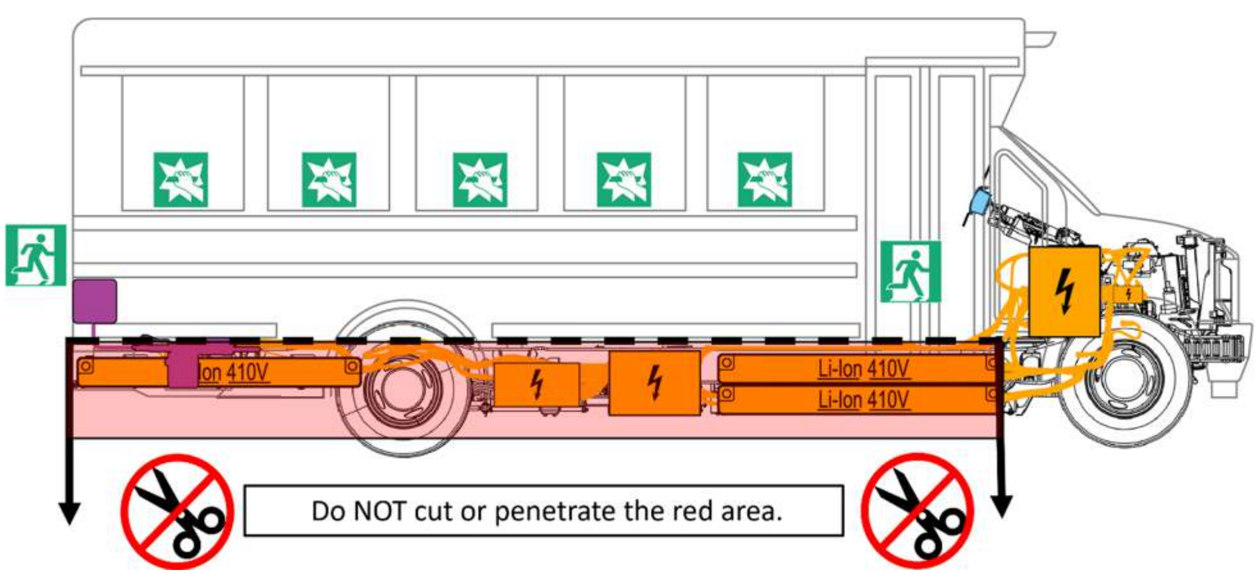


This school bus has an airbag for the driver.



No-Cut Zones

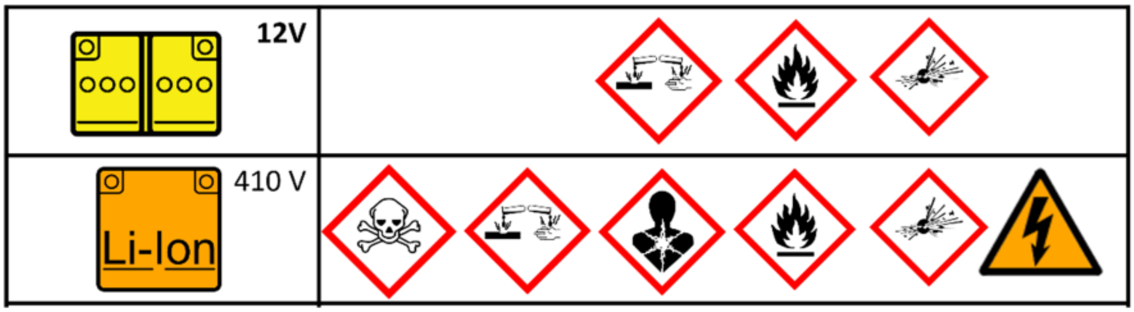
Occupants can evacuate through the front door, emergency exit window, or the back emergency exit. They may also be accessed through breaking the window(s).



WARNING: If occupant removal is necessary, do not cut any of the high-voltage lines under vehicle (all high-voltage lines are orange).

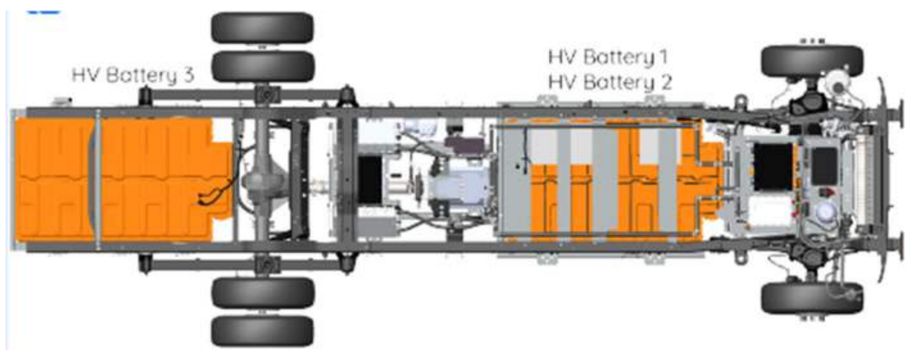
5. Stored energy / liquids / gases / solids

Electric Components



High-Voltage Battery Packs

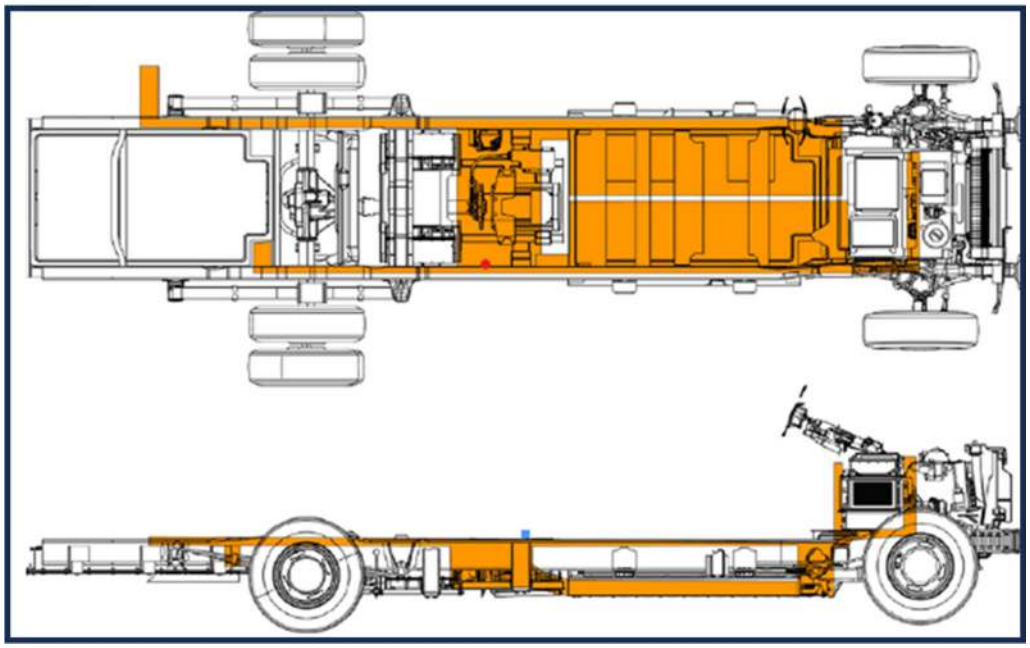
The Van-Con is equipped with 350V lithium-ion, high-voltage batteries mounted between and below the chassis frame rails. The batteries are made up of multiple cells inside an enclosed housing that are cooled using R-134 refrigerant.



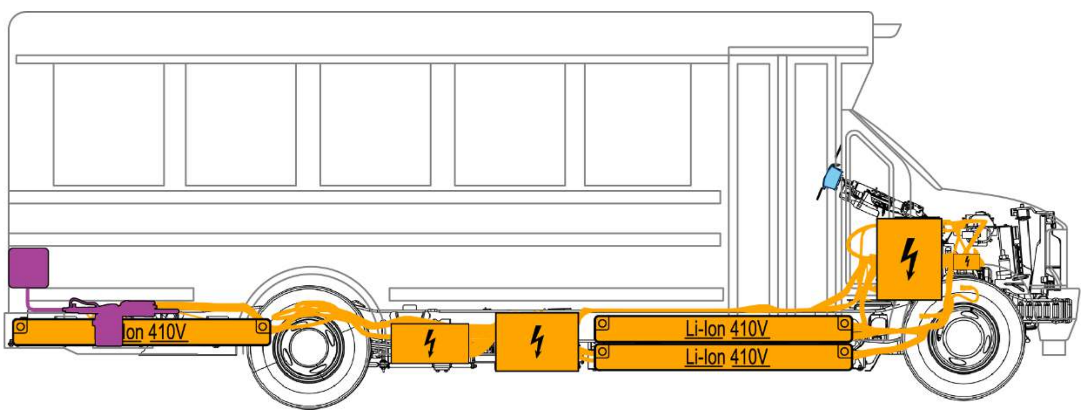
WARNING: Use caution to avoid damaging the high-voltage batteries and battery assembly cover during vehicle rescue and recovery operations. The batteries may have stored energy within them and present a danger to personnel.

High-Voltage Cables

High-voltage cables are shown in orange. There are high-voltage cables that run the length of the vehicle between and under the chassis frame rails. High-voltage cables are colored orange and protected by orange looming.

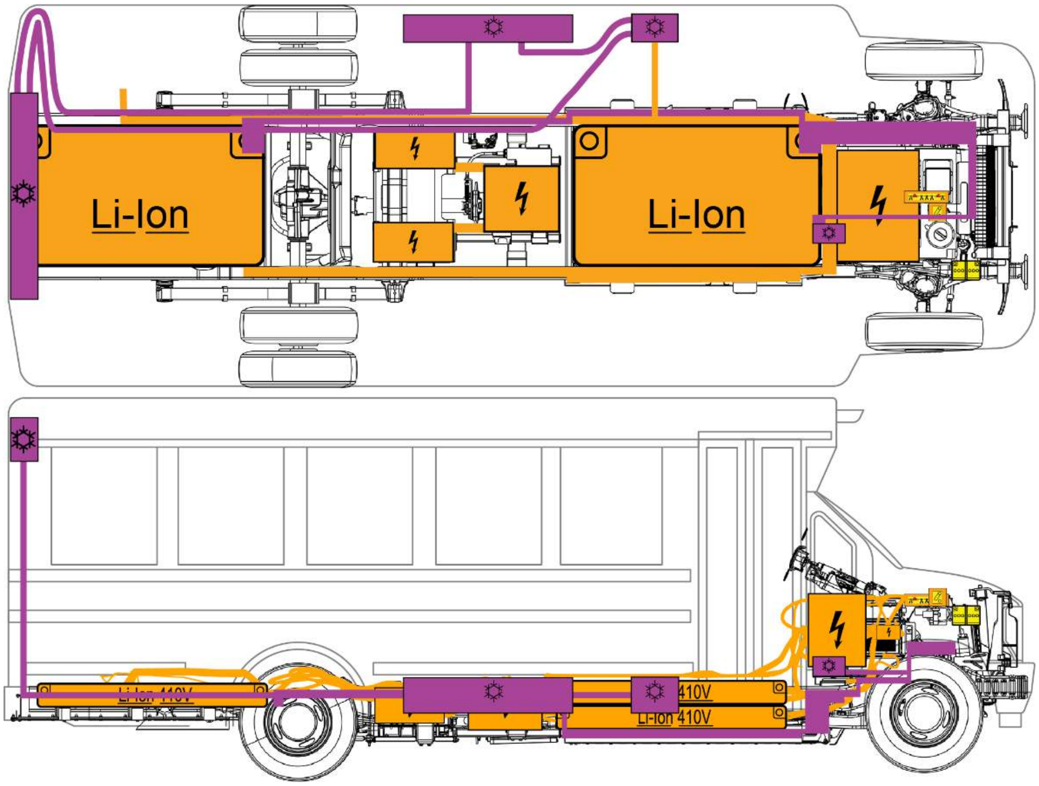


The heating system (in purple) is also high-voltage.



R134 Refrigerant System

Refrigerant lines are shown below in purple. There are refrigerant lines that run the length of the vehicle between, under, and outside of the chassis frame rails. Refrigerant lines are a combination of black flexible hose lines and welded metal lines.



WARNING: Do not compromise the refrigerant lines with rescue tools.

6. In Case of Fire

Firefighting

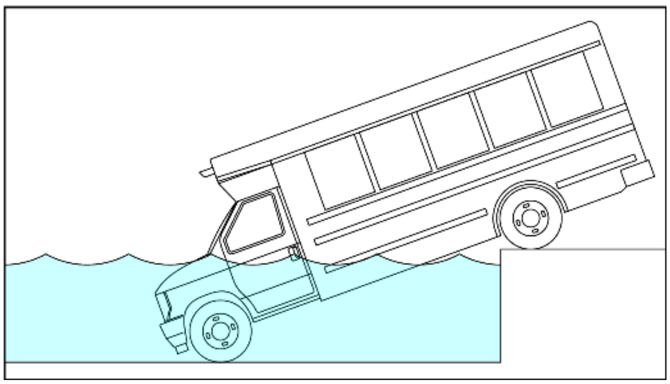
Use water to fight a high-voltage battery fire. Always obtain an additional water supply early, it can take thousands of gallons of water, applied directly to the battery, to fully extinguish and cool down a battery fire.

Battery fires can take up to 24 hours to fully cool. There must be no fire, smoke, or heating present in the high-voltage batteries for at least 60 minutes. The batteries must be completely cooled before the vehicle can be released to law enforcement, vehicle transportation, or any other second responder.

WARNING: Damaged lithium-ion batteries may re-ignite after the fire is extinguished. Use an infrared camera to monitor the temperature of the battery.

WARNING: A burning high-voltage battery releases heated gases and toxic vapors. Always wear full personal protective equipment (PPE), including a self-contained breathing apparatus (SCBA).

7. In case of Submersion

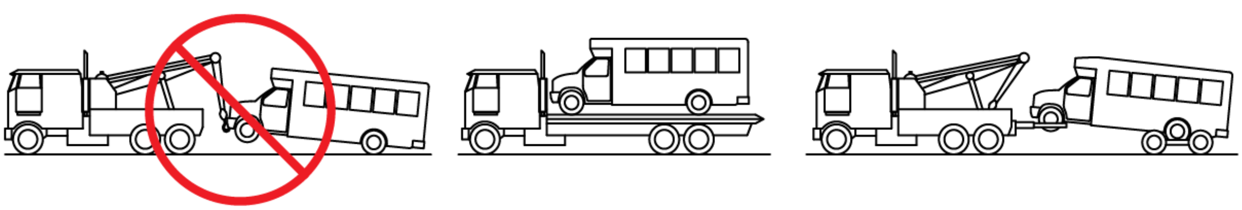


Remove the vehicle from the water and allow the water to drain before disabling the high-voltage system.

Follow Section 3 Disable Direct Hazards / Safety Regulations.

WARNING: Vehicles that have been submerged in water have a greater risk of a high-voltage battery fire. First responders should be prepared to respond to a high-voltage battery fire.

8. Towing / Transportation / Storage



When towing the school bus, never tow it with the front or rear wheels on the ground. Always use dollies or use a flatbed truck.

WARNING: Even with the powertrain shut off, magnets in traction motors generate dangerous high-voltage when the vehicle is towed with wheels on the ground or coasting.

WARNING: Always tow the vehicle on a cradle or with dollies when moving it or disconnect the driveshaft and remove the rear section of the driveshaft from the vehicle.

WARNING: A vehicle involved in a submersion, fire, or collision that has compromised the high-voltage batteries should be stored in an open area at least 50 feet from any other vehicle, buildings, or property.

9. Important Additional Information

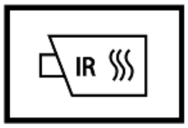
WARNING: Only first responders who are trained and equipped at the technicial level per the National Fire Protection Association (NFPA) and are familiar with the lift points should lift or manipulate the vehicle.

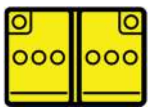
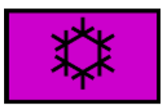
WARNING: Even when the red emergency cut-off switch is turned OFF to disconnect the high-voltage and the 12V, some high-voltage and 12V components may contain stored electricity.

WARNING: If there are leaking fluids, sparks, smoke, flames, increased temperature, gurgling, popping or hissing noises from the high-voltage compartment, ventilate the area and call emergency responders.

WARNING: A burning high-voltage battery releases heated gases and toxic vapors. Always wear full personal protective equipment (PPE), including a self-contained breathing apparatus (SCBA).

10. Explanation of Pictograms

	Infrared (IR) device also known as a thermal imaging camera (TIC)
	General warning
	Electrical warning
	Flammable
	Explosive
	Corrosive substances
	Hazardous to human health
	Acute toxicity
	Use water to extinguish fire
	Electric vehicle
	High-voltage battery

	High-voltage cable cut
	High-voltage component
	High-voltage cable/component
	Emergency disconnect switch
	12V battery
	Cooling component
	Cooling line
	Emergency door opener
	Break to obtain access
	Emergency exit